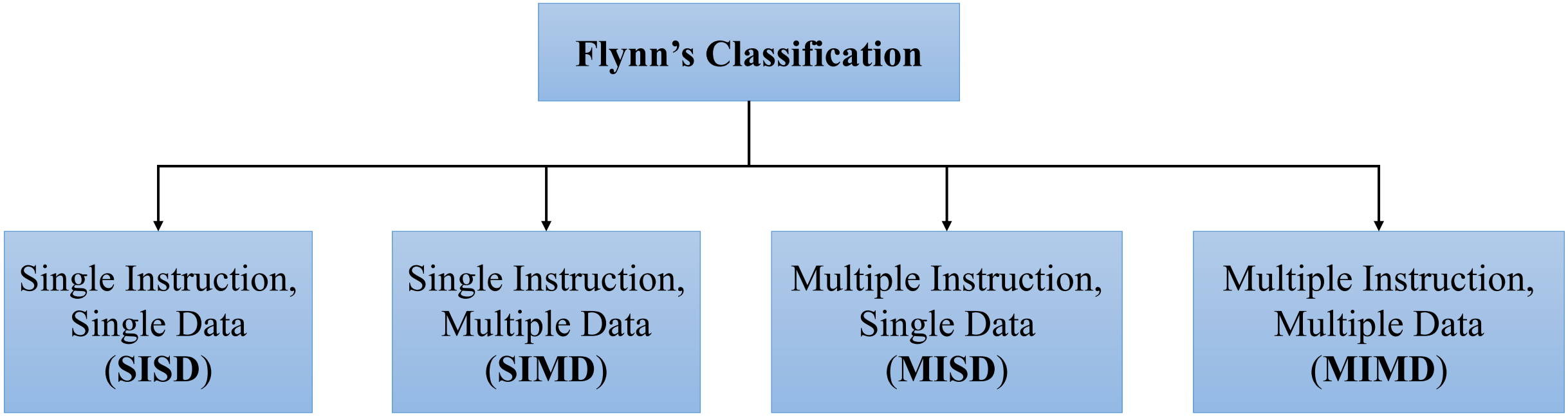
Flynn's Architectural Classifications

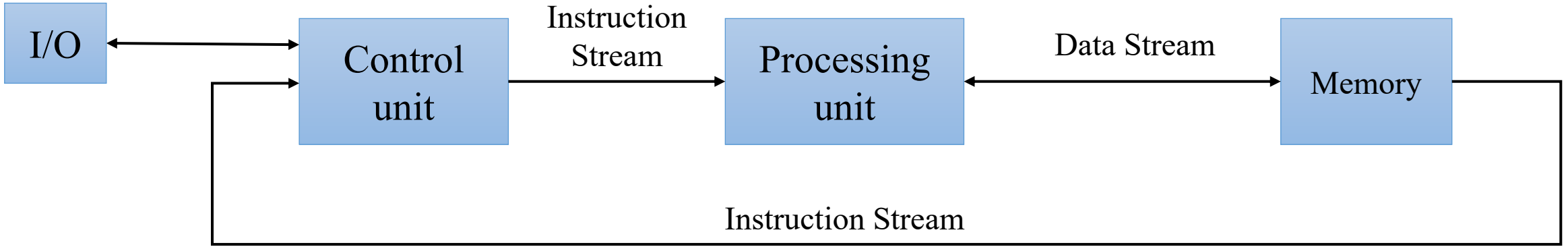
CSE Sem-4

- In 1972, Michael J Flynn introduced a scheme for classifying computer architecture with the concept of **instruction stream** and **data stream**.
- **Instruction stream** is a sequence of instructions as executed by the machine.
- **Data stream** is defined as a sequence of data including input, partial or temporary results which are called by the instruction stream.



Single Instruction Stream and Single Data Stream

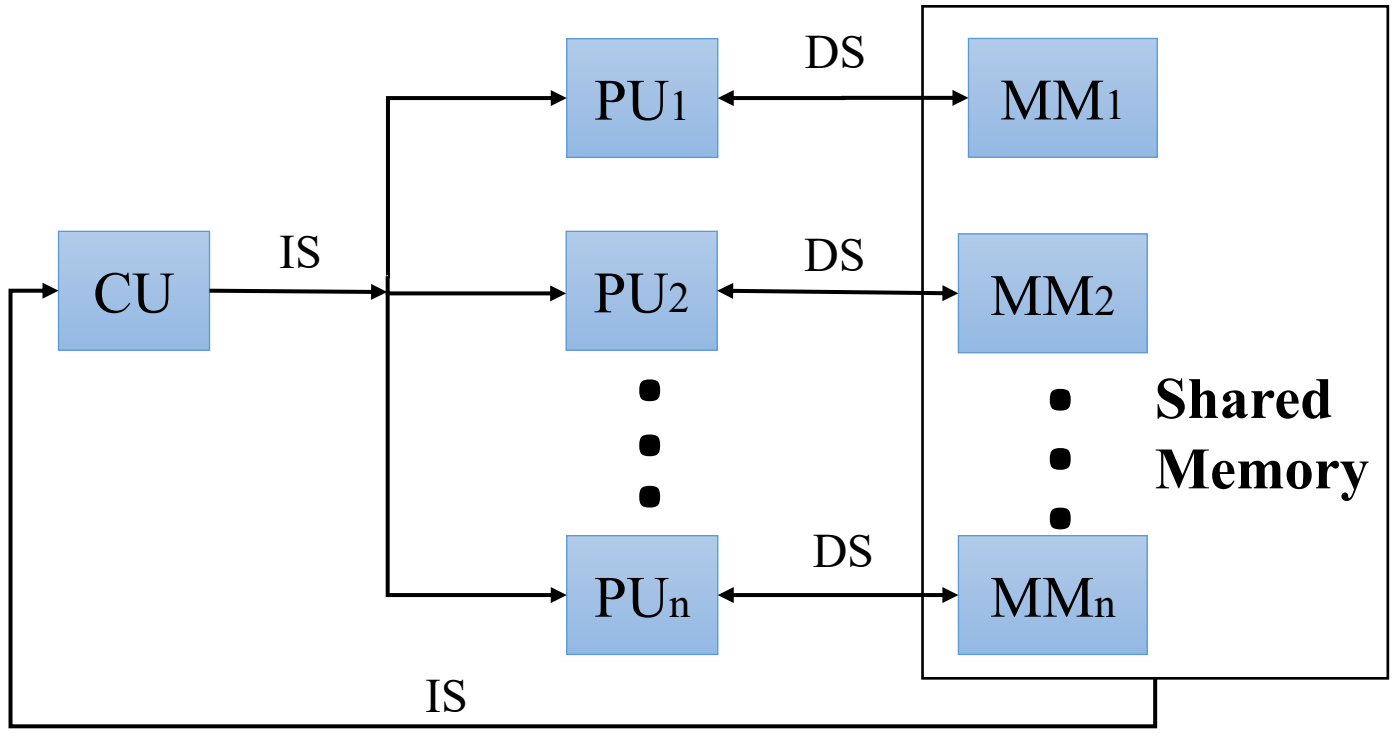
- Represents the organization of a **Uniprocessor** containing a control unit, a processor unit and a memory unit. **Only one instruction is executed at a time**.
- Instructions are executed sequentially, and the system may or may not have parallel processing capabilities.
- Parallel processing may be achieved by pipeline processing.



- Instructions are decoded by the control unit and sends the instructions to the processing units for execution.
- Data Stream flows between the processors and memory bi-directionally.
- Example: Uniprocessor computers, minicomputers

Single Instruction Stream and Multiple Data Stream

- Represents an organization that includes **many processing units** under the supervision of a **common control unit.**
- All processors receive the same instruction from the control unit but operate on different data.
- The shared memory unit must contain multiple modules so that it can communicate with all the processors simultaneously.



- Used in array processing.
- For example, we have 2 arrays –

  $X = [X_0, X_1, X_2, X_3]$

  $Y = [Y_0, Y_1, Y_2, Y_3]$

  We have to perform X + Y

  So, $PU_1 = X_0 + Y_0$

  $PU_2 = X_1 + Y_1$

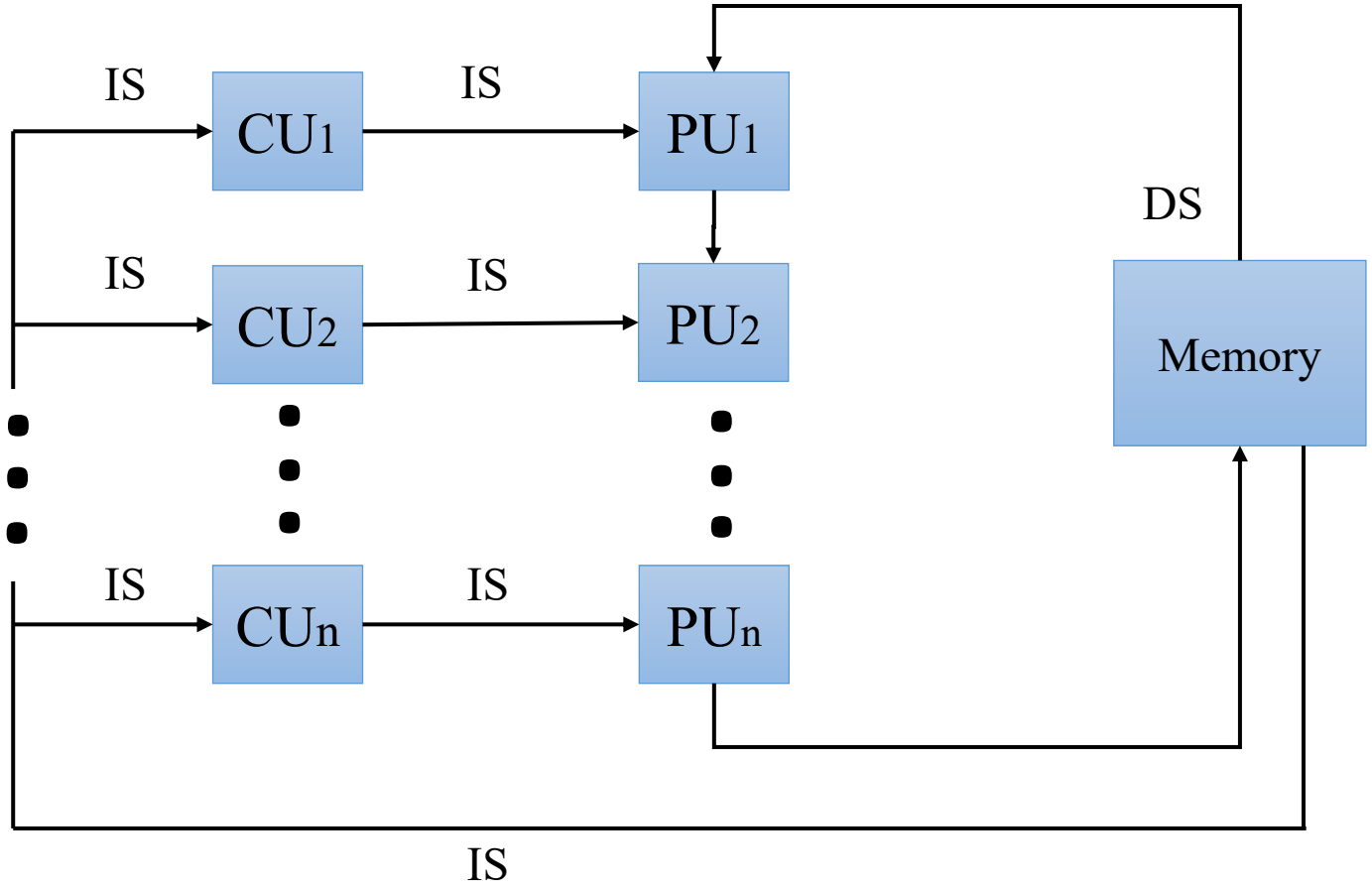
  $PU_3 = X_2 + Y_2$

  $PU_4 = X_3 + Y_3$

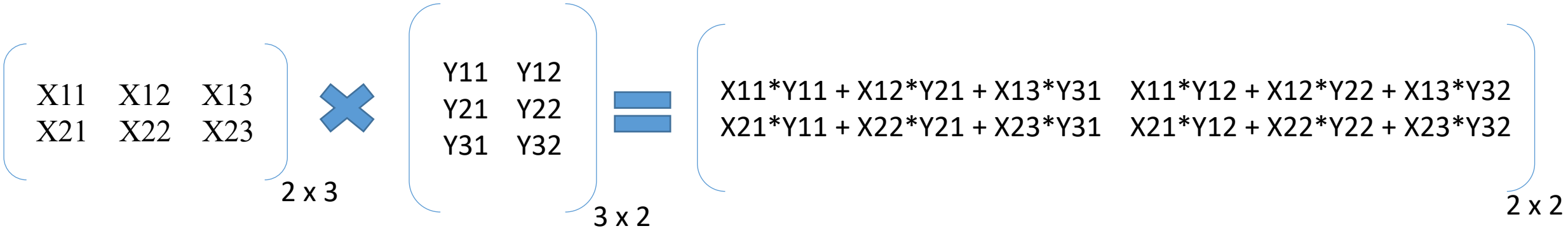
So, a single instruction (addition) performing on different data elements.

Multiple Instruction Stream and Single Data Stream

- **Multiple processing units** operate on **one single-data stream**. Same data stream flows through a linear array of processors executing different instruction streams.
- Each processing unit operates on the data independently via separate instruction stream.

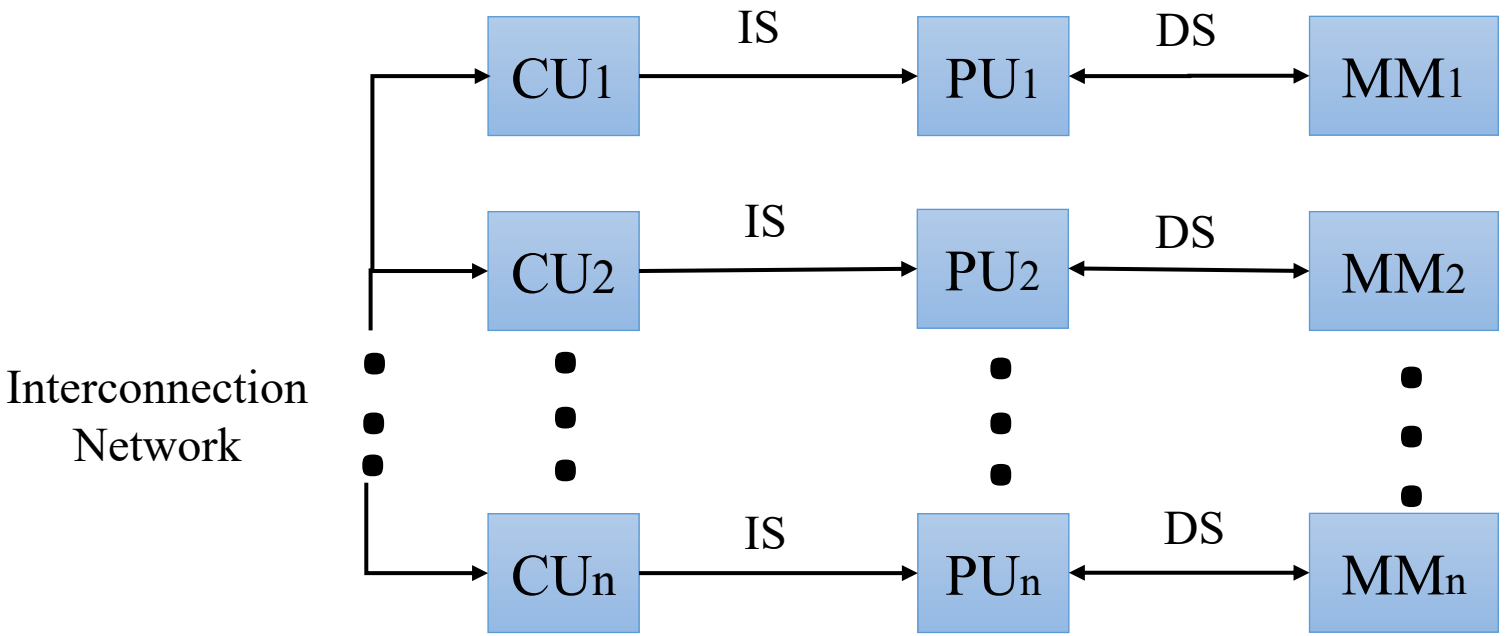


- Used in **systolic arrays**. A systolic array is a pipelined structure containing n processing units. Each unit independently computes a partial result with the data received from its previous PU, store the result and passes to its next PU. So, it is a synchronous communication.
- Used in parallel integration, matrix multiplication, data sorting.
- MISD structure **is only of theoretical interest** since no practical system has been constructed using this organization.


X11 X12 X13
X21 X22 X23
2 x 3
Y11 Y12
Y21 Y22
Y31 Y32
3 x 2
X11*Y11 + X12*Y21 + X13*Y31 X11*Y12 + X12*Y22 + X13*Y32
X21*Y11 + X22*Y21 + X23*Y31 X21*Y12 + X22*Y22 + X23*Y32
2 x 2

Multiple Instruction Stream and Multiple Data Stream

- In this organization, all processors in a parallel computer can execute different instructions and operate on various data at the same time.
- Each processor has a separate program and an instruction stream is generated from each program.



- Multicomputer System